

IBM HR Analytics Employee Attrition Prediction

Foundation Notes: KPIs, Charts, Scaling, Evaluation Metrics, Models, and
Business Interpretation

Prepared for: Ruturaj Mokashi

Final selected model: Random Forest with threshold 0.25

Purpose of this guide

This document is designed to build foundation-level clarity from your HR attrition project. It explains the full workflow, the calculations behind KPIs, how charts were built, what scalers do, how evaluation metrics work, why models were chosen, and how the final model was selected.

Design note: The visual style is IBM-inspired, using dark neutral panels and blue/cyan accents. No official IBM logo or endorsement is used.

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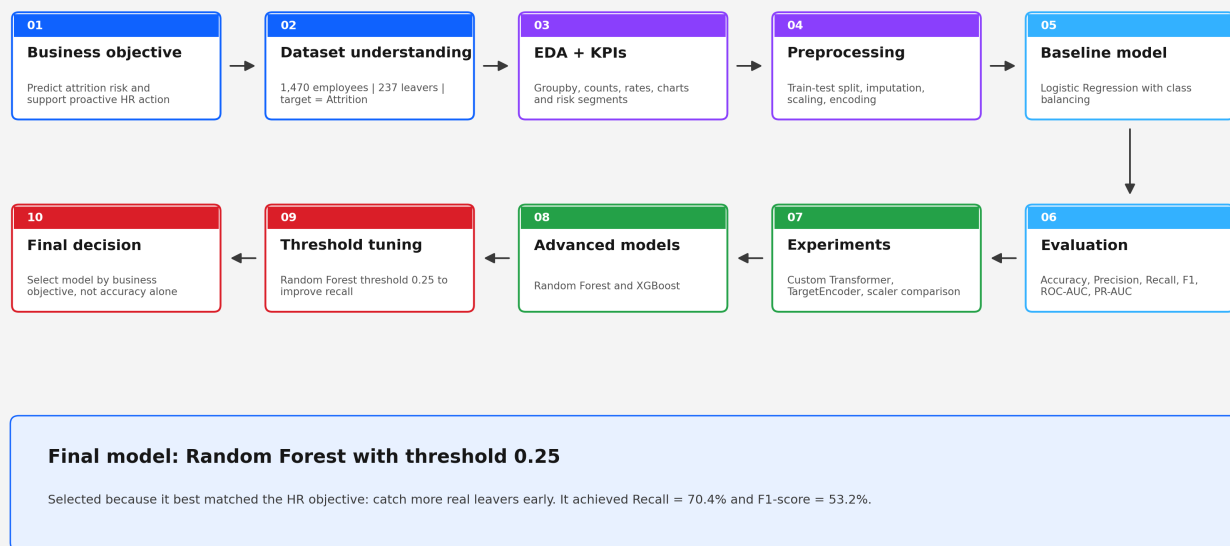
SECTION 01

Big-picture workflow

The full project journey in one clean visual.

Employee Attrition Prediction - Project Workflow

IBM-inspired foundation guide | From business question to HR early-warning model



Prepared for Rituraj Mokashi

Figure 1. Project workflow from business problem to final HR recommendation.

Key idea

We did not build charts and models randomly. Each step served a business purpose: understand risk, prepare data safely, train models, evaluate trade-offs, and select the model that best supports HR action.

SECTION 02

Dataset and business objective

What the data represents and why the project exists.

Dataset recap

Item	Value	Meaning
Dataset	IBM HR Analytics Employee Attrition	Employee-level HR dataset used for attrition prediction practice
Rows	1,470	One row represents one employee
Target column	Attrition	The answer we want to predict: Stayed or Left
Employees stayed	1,233	Majority class
Employees left	237	Minority class
Attrition rate	16.12%	237 / 1,470 employees left

Business objective

Identify which employees are more likely to leave and create an early-warning model that helps HR take proactive action.

- Understand high-risk employee segments using EDA.
- Build a model that predicts attrition risk from employee information.
- Select a model based on HR usefulness, not accuracy alone.
- Use predictions to support ethical HR conversations, not replace human judgment.

SECTION 03

KPI calculations, groupby, joins and aggregation

How the numbers behind your charts were calculated.

The core pattern

Most KPIs used the same pattern: group employees by a column, count employees, count leavers, then divide leavers by total employees in that group.

Core attrition-rate formula

Metric	Formula	Project example
Total employees	count of rows	1,470
Employees left	sum of Attrition = 1	237
Overall attrition rate	employees left / total employees	$237 / 1,470 = 16.12\%$
Group attrition rate	leavers in group / employees in group	OverTime Yes: $127 / 416 = 30.5\%$

How groupby works

Step	Plain English	Python idea
1	Choose a grouping column	<code>groupby("OverTime")</code>
2	Count employees in each group	<code>size()</code> or <code>count()</code>
3	Count leavers in each group	<code>sum Attrition where Left = 1</code>
4	Calculate rate	<code>leavers / employees * 100</code>
5	Sort or plot	<code>sort_values()</code> or chart

Example: overtime KPI calculation

OverTime group	Employees	Leavers	Attrition rate
Yes	416	127	$127 / 416 = 30.5\%$
No	1,054	110	$110 / 1,054 = 10.4\%$

What about joins?

This project mainly used groupby and aggregation. A join would be needed if employee data lived in separate tables.
Example: salary table + survey table joined using EmployeeNumber.

SECTION 04

Charts and visual thinking

What each chart type means and how to read it.

Chart selection map

Question	Chart type	What it answers
How many?	Bar chart	How many employees are in each group?
What rate?	Horizontal bar or dot plot	Which group has higher attrition?
Distribution?	Box plot or violin plot	How are numeric values spread out?
Two dimensions?	Bubble/combo chart	How do size and risk compare?
Model quality?	Confusion matrix, PR curve, ROC curve	How well did the model predict?

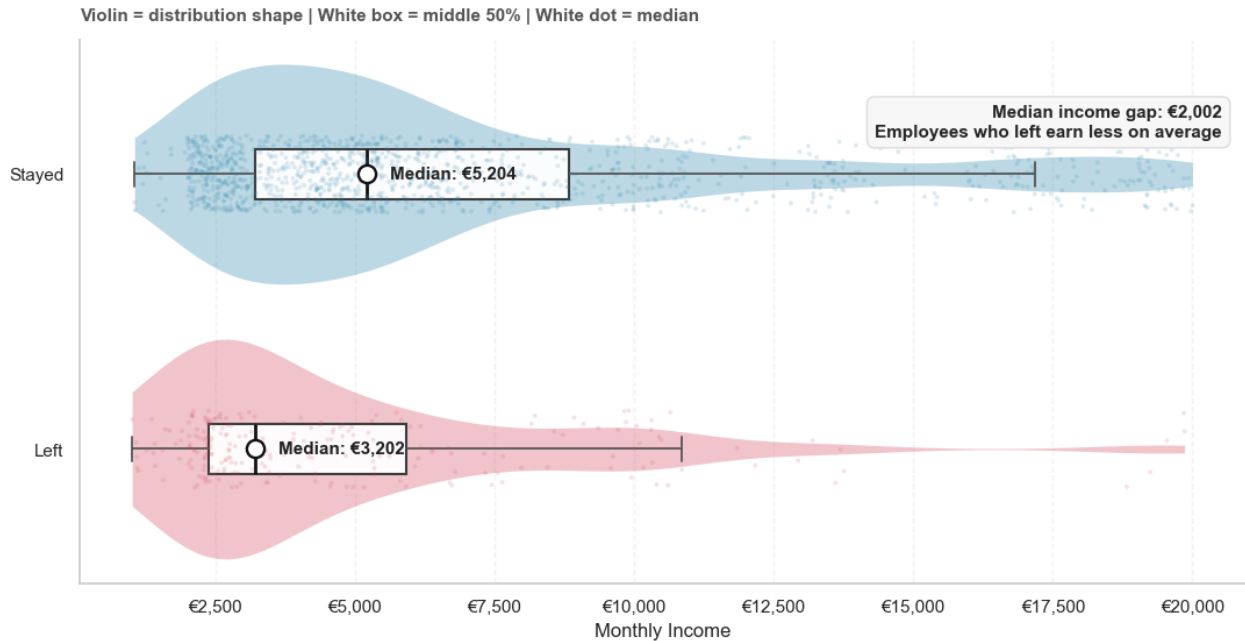
Box plot: the mysterious parts explained

Element	Meaning	How to read it
Median line	Middle value	50% of employees are below it and 50% are above it
Box	Middle 50% of values	The normal central range of the group
Lower edge	25th percentile	25% of values are below this
Upper edge	75th percentile	75% of values are below this
Whiskers	Usual lower and upper range	How far typical values stretch
Dots	Individual employees or outliers	Each dot is a real observation

Violin plot

A violin plot adds shape to a box plot. Wider sections mean more employees are concentrated there. Narrow sections mean fewer employees.

Monthly Income Distribution by Attrition Status



Income distribution: employees who left tend to have lower monthly income.

SECTION 05

Scaling explained clearly

StandardScaler, MinMaxScaler and RobustScaler without mystery.

Scaler Cheat Sheet

StandardScaler

Centers around 0

Uses mean and standard deviation

Best common default for Logistic Regression



MinMaxScaler

Squeezes into 0 to 1

Uses min and max

Best performer in this project for LogReg



RobustScaler

Uses the middle of the data

Uses median and IQR

Helpful when outliers are strong

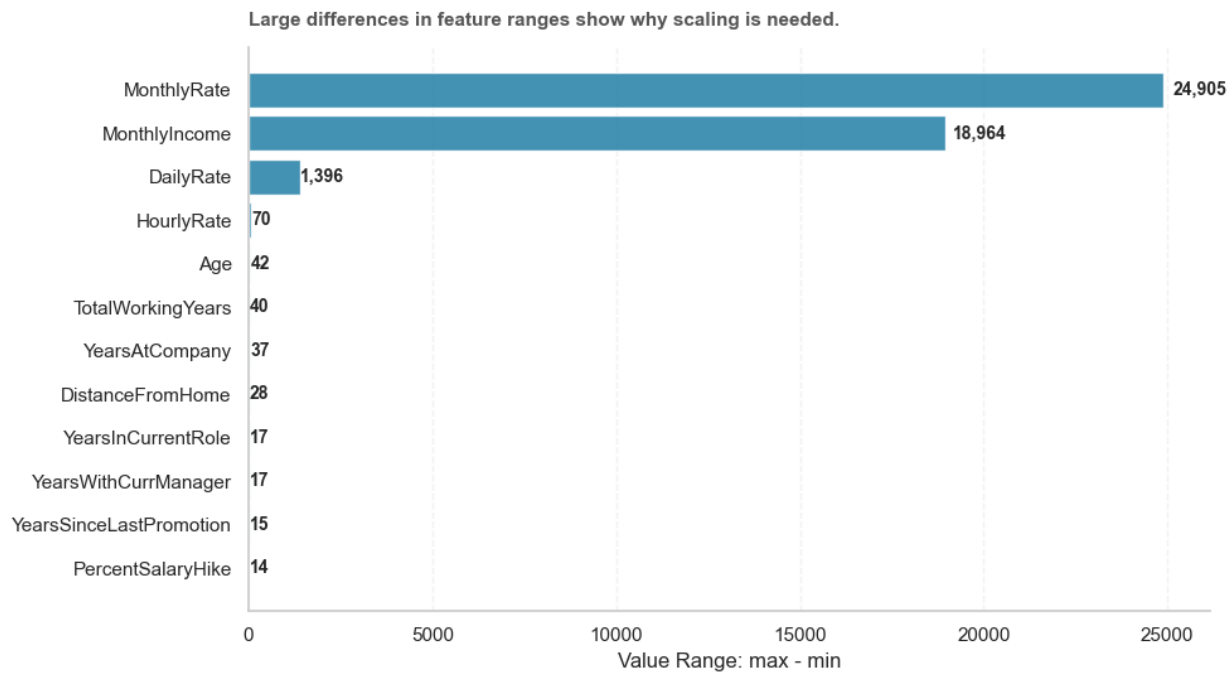


Scaler cheat sheet.

Why scaling was needed

Some numeric columns had huge values while others had tiny values. For example, MonthlyIncome can be close to 20,000, while JobSatisfaction is only 1 to 4. Models such as Logistic Regression can be affected by these scale differences.

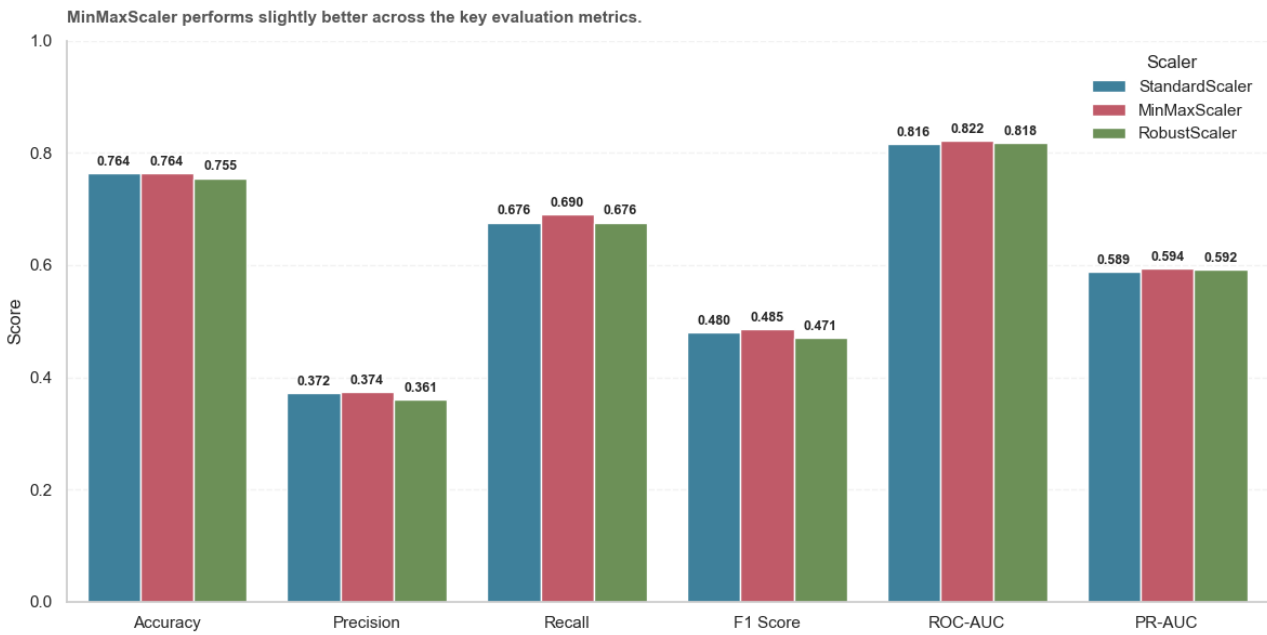
Numeric Feature Ranges Before Scaling



Numeric feature ranges before scaling.

Scaler	What it does	Simple mental picture
StandardScaler	Uses mean and standard deviation	0 = average; positive = above average; negative = below average
MinMaxScaler	Uses min and max	0 = smallest value; 1 = largest value
RobustScaler	Uses median and IQR	Less sensitive to extreme values/outliers

Scaler Comparison for Logistic Regression



Scaler comparison: MinMaxScaler performed slightly better for Logistic Regression in this project.

SECTION 06

Train-test split and prediction logic

Why it is still prediction even when the answers exist.

Total data: 1470 employees

Training set: 1029 employees
Model learns here

Test set: 441 employees
Answers hidden first

Prediction is fair because the model is tested on employees whose Attrition answer was hidden during prediction.

Train-test split logic.

Exam analogy

Training data is practice questions with answers. Test data is exam questions. The teacher has the answer sheet, but the model does not see it before predicting.

Dataset part	Rows	Purpose
Training set	1,029	Model learns patterns
Test set	441	Model is evaluated on unseen examples

The test answers already exist in the dataset, but they are hidden from the model during prediction. The model sees employee features such as age, job role, income and overtime, but not the true Attrition value. After predictions are made, we compare the model output with the hidden answers.

SECTION 07

Evaluation metrics without jargon

Accuracy, Precision, Recall, F1, ROC-AUC and PR-AUC.

Evaluation Metrics - Plain-English Map

Accuracy

All correct predictions / all employees
Good general view, but risky with imbalanced data

Precision

Of predicted leavers, how many really left?
Controls false alarms

Recall

Of real leavers, how many did we catch?
Critical for HR early-warning

F1 Score

Balance between precision and recall
Best single metric for this use case

ROC-AUC

Can the model rank leavers above stayers?
Probability ranking quality

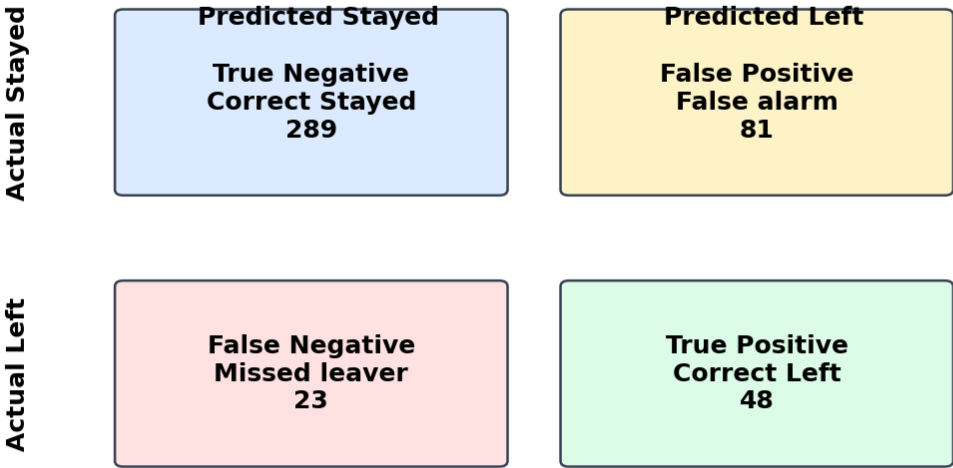
PR-AUC

Precision-recall quality across thresholds
Useful for imbalanced data

Evaluation metrics in plain English.

Confusion matrix foundation

Confusion Matrix: How predictions are checked



$Recall = TP / (TP + FN) = 48 / (48 + 23) = 67.6\%$
 $Precision = TP / (TP + FP) = 48 / (48 + 81) = 37.2\%$

How Accuracy, Precision, Recall and F1 come from the confusion matrix.

Our model confusion matrix

Confusion Matrix — Baseline Logistic Regression

Shows correct predictions and model errors on the test set.



Confusion matrix for baseline Logistic Regression.

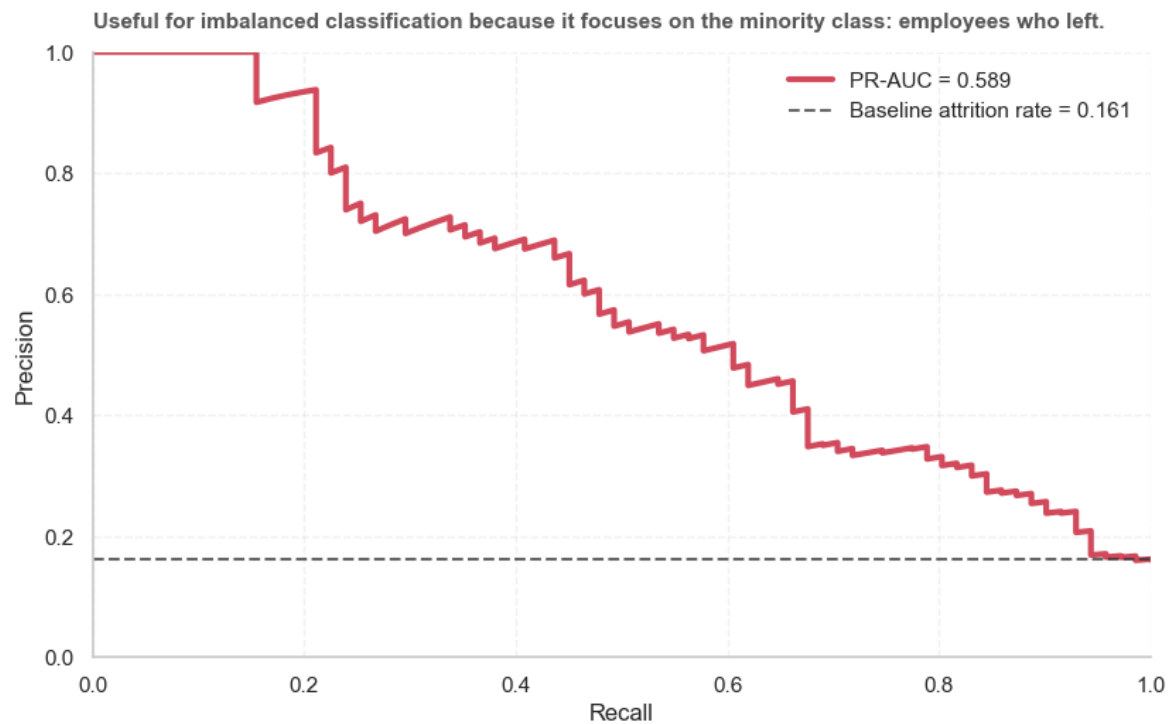
Most important metric for HR

Recall matters because missed leavers are risky. If an employee actually leaves but the model predicts Stayed, HR loses the chance to intervene.

ROC-AUC and PR-AUC

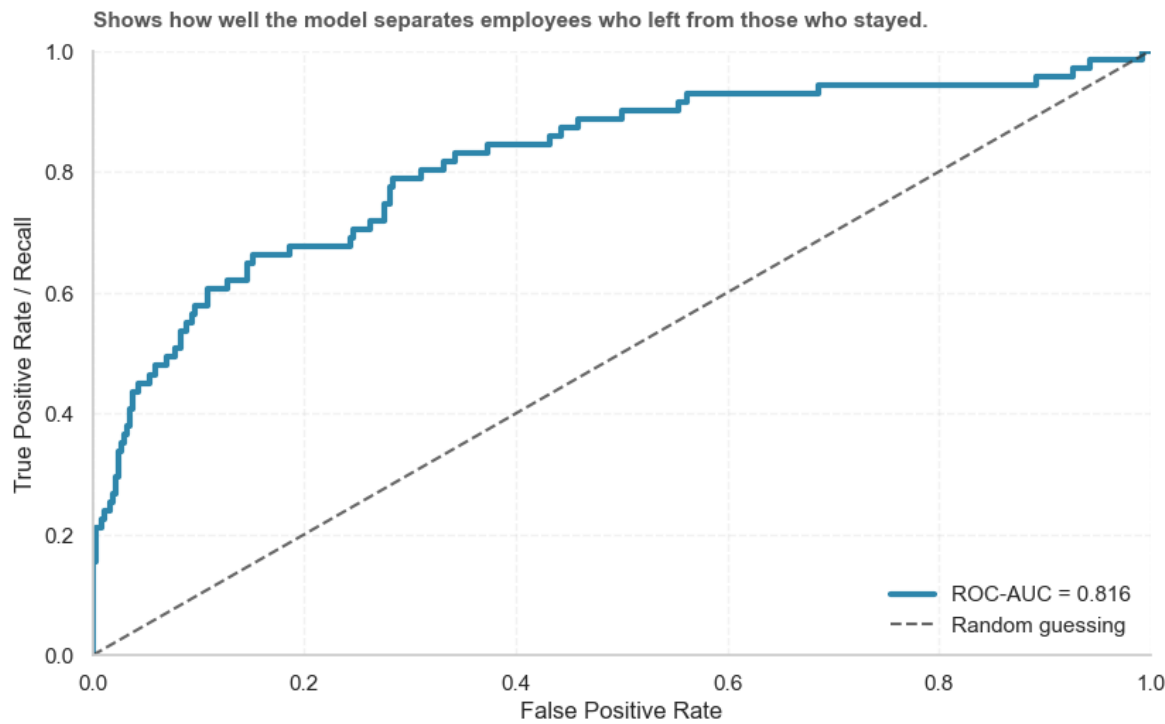
Metric	Plain English	Why useful
ROC-AUC	How well the model ranks leavers above stayers	Good overall separation view
PR-AUC	Precision-recall quality across thresholds	Useful for imbalanced datasets

Precision-Recall Curve — Baseline Model



Precision-Recall curve: useful because the leaver class is small.

ROC Curve — Baseline Model



ROC curve: overall separation ability.

SECTION 08

Models used in this project

What the models do and why we chose them.

Logistic Regression

What it does

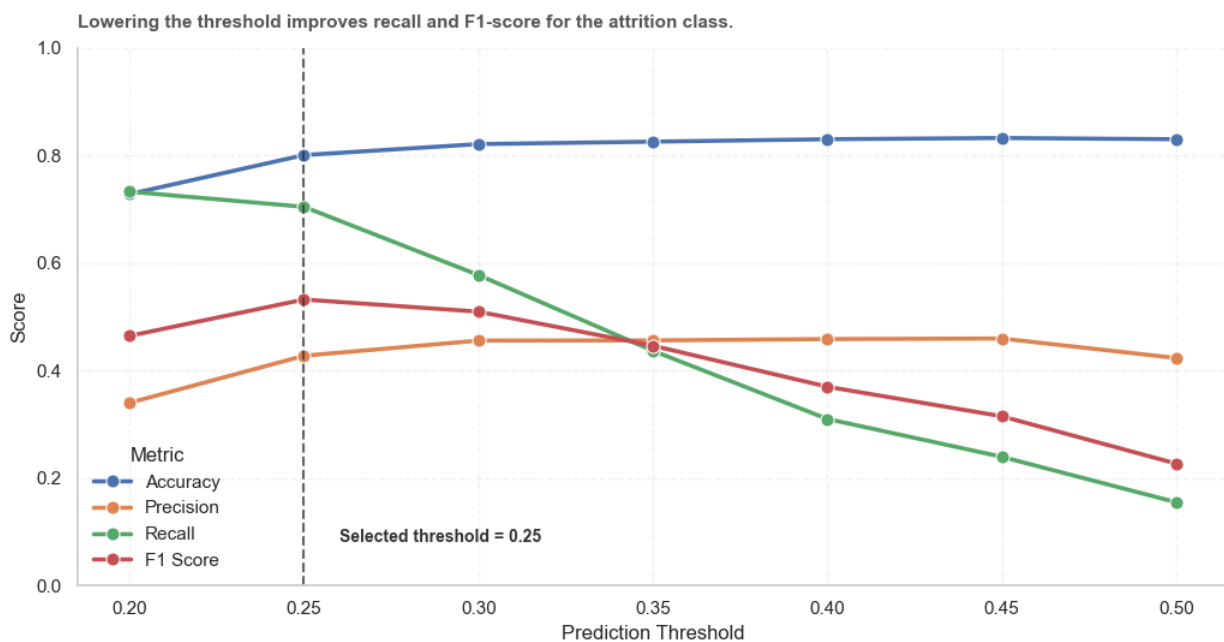
A classification model that estimates the probability that an employee belongs to the Left class. It is a strong baseline and easy to interpret using coefficients.

Random Forest

What it does

Builds many decision trees and lets them vote. It can capture more complex relationships and does not require feature scaling.

Random Forest Threshold Tuning



Threshold tuning made Random Forest much better for the HR use case.

XGBoost

What it does

Builds trees sequentially. Each new tree tries to fix mistakes made by previous trees. It is widely used for high-performance tabular ML.

Custom Transformer and TargetEncoder

Technique	Purpose	Result
Custom Transformer	Created JobSatisfaction_x_YearsAtCompany	Implemented successfully, but did not meaningfully improve performance
TargetEncoder	Encoded categories using target-based values	Useful concept, but did not beat OneHotEncoder baseline

Leakage warning

TargetEncoder uses the target column. It must be inside the pipeline so it learns encodings only from training data, not from test data.

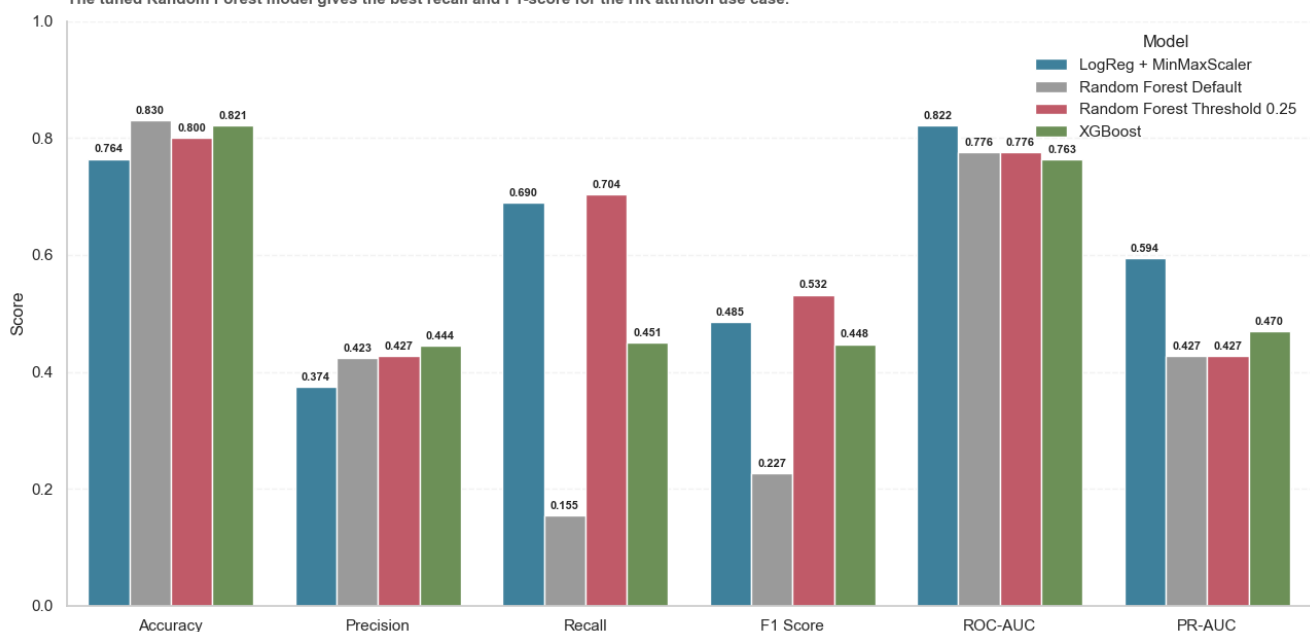
SECTION 09

Final model comparison and decision

How we selected the final model using the business objective.

Final Model Comparison

The tuned Random Forest model gives the best recall and F1-score for the HR attrition use case.



Final model comparison.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
LogReg + MinMaxScaler	76.4%	37.4%	69.0%	48.5%	82.2%	59.4%
Random Forest Default	83.0%	42.3%	15.5%	22.7%	77.6%	42.7%
Random Forest Threshold 0.25	80.0%	42.7%	70.4%	53.2%	77.6%	42.7%
XGBoost	82.1%	44.4%	45.1%	44.8%	76.3%	47.0%

Final model selected

Random Forest with threshold 0.25 was selected because it achieved the best recall and F1-score for the HR early-warning objective.

The final model was not chosen because it had the highest accuracy. It was chosen because it best matched the business goal: identify employees who may actually leave early enough for HR to take action.

SECTION 10

Industry model overview

Where these models fit in modern data teams.

Model family	Examples	Common industry use
Linear models	Logistic Regression, Linear Regression	Baselines, interpretable risk scoring
Tree ensembles	Random Forest, ExtraTrees	Tabular business prediction and feature importance
Gradient boosting	XGBoost, LightGBM, CatBoost	High-performance churn, fraud, pricing and risk models
Neural networks	MLP, deep learning models	Images, recommendations and large-scale data
Transformer models	BERT, GPT-style models	Text understanding, summarization and search
AutoML platforms	H2O, DataRobot, AutoGluon	Fast model benchmarking and production workflows

Professional mindset

A model is not selected because it sounds advanced. It is selected because it performs best for the business objective, is maintainable, and can be explained to stakeholders.

SECTION 11

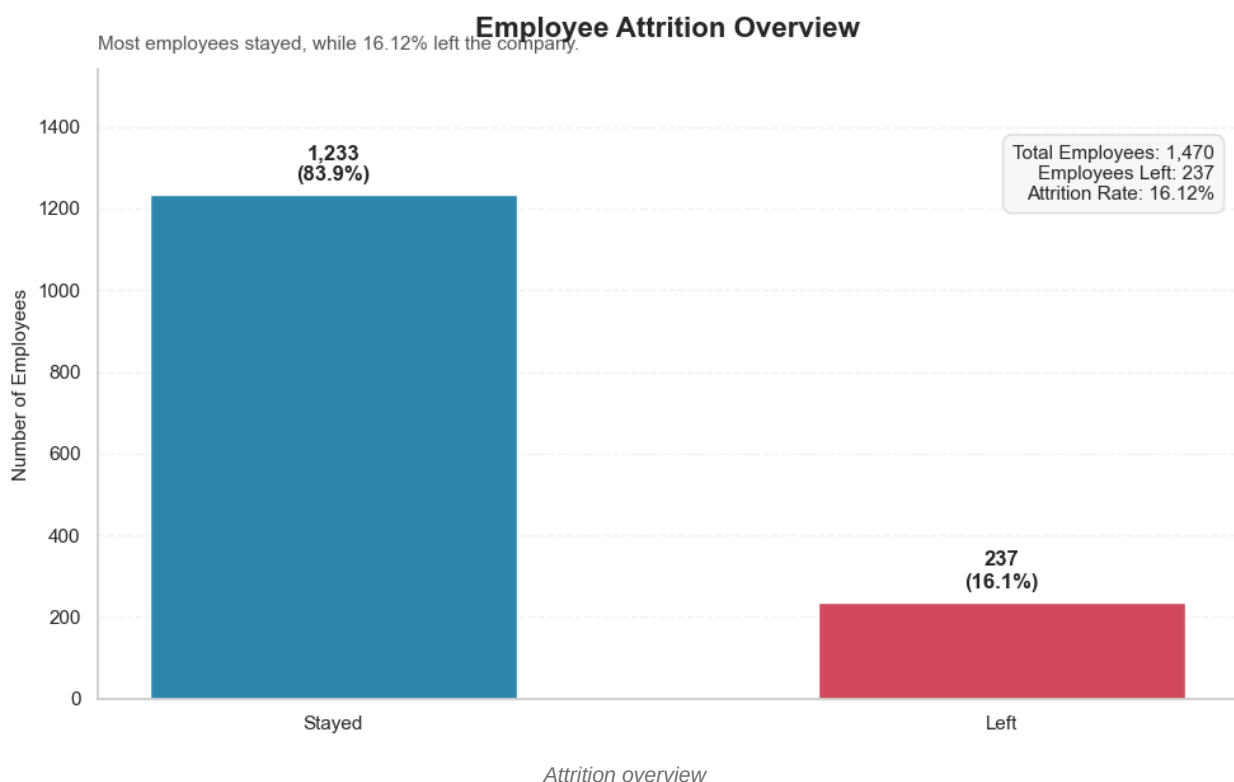
Chart appendix and study notes

All major charts from the project and what each one teaches.

Attrition overview

How it was built

Counts stayed vs left and sets the imbalance context. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

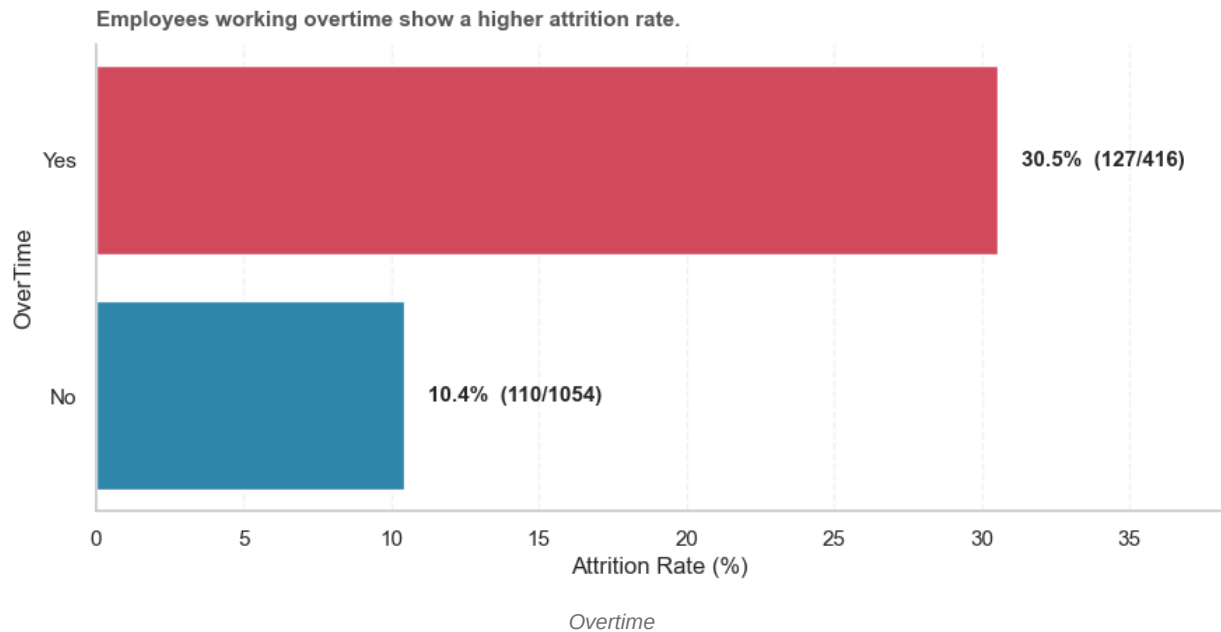


Overtime

How it was built

Shows overtime employees have much higher attrition. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Attrition Rate by Overtime Status

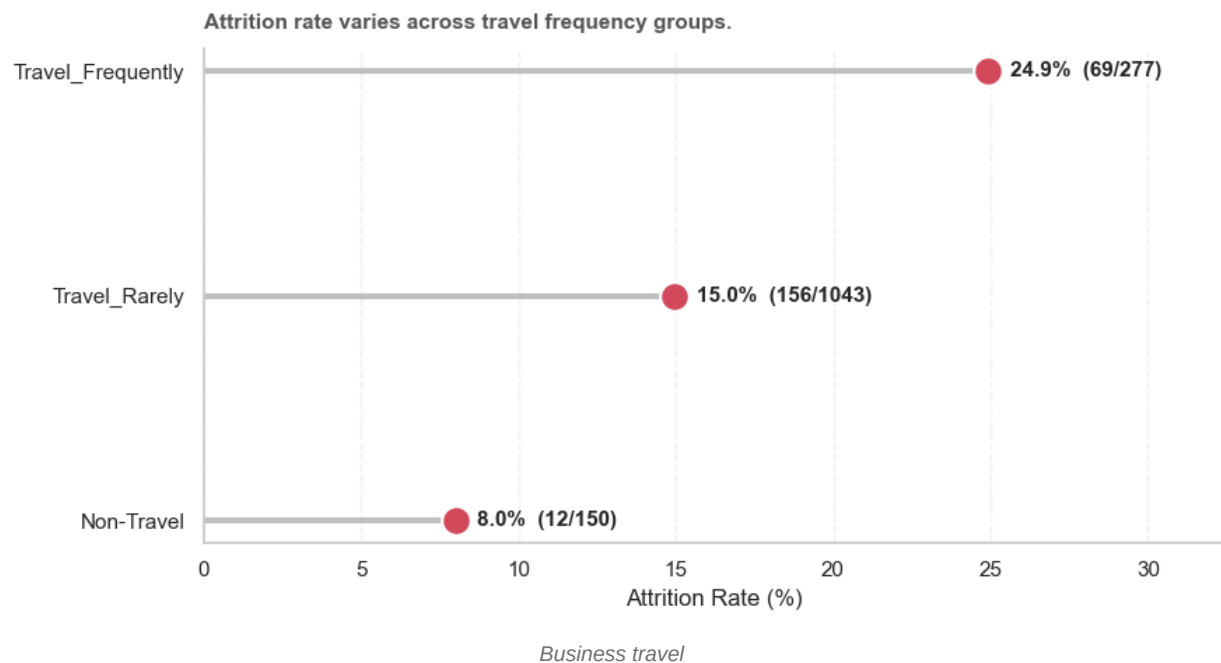


Business travel

How it was built

Shows frequent travelers have higher attrition. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Attrition Rate by Business Travel Frequency

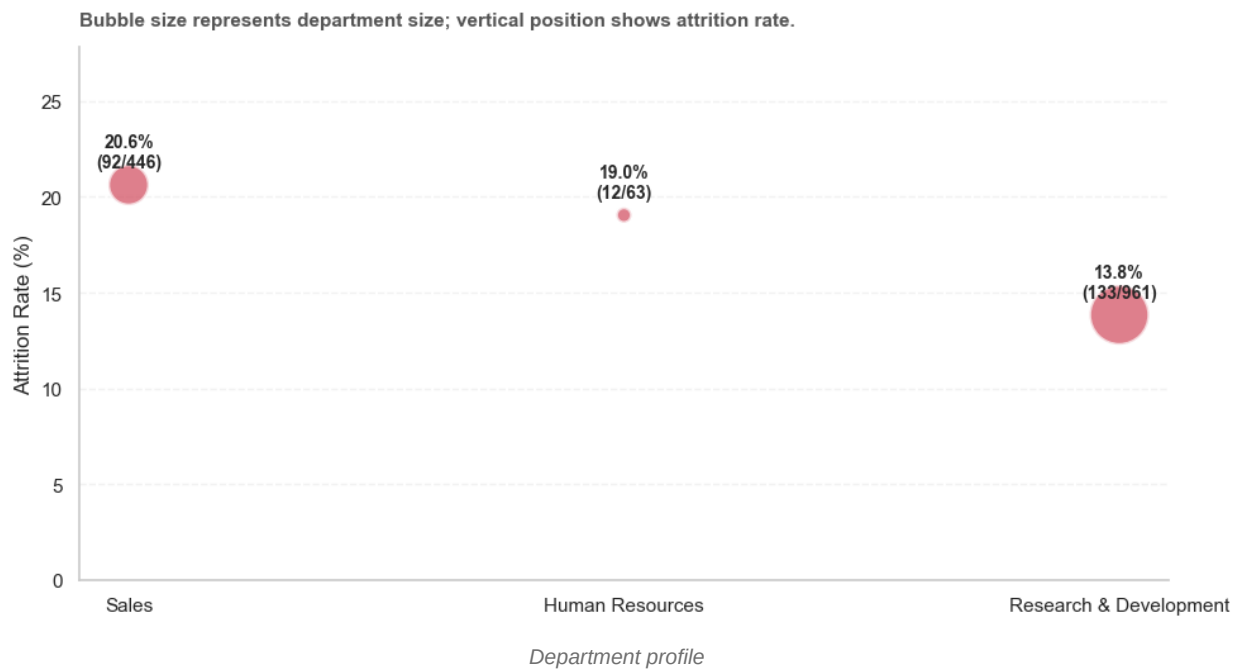


Department profile

How it was built

Compares department size and attrition rate. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Department Attrition Profile

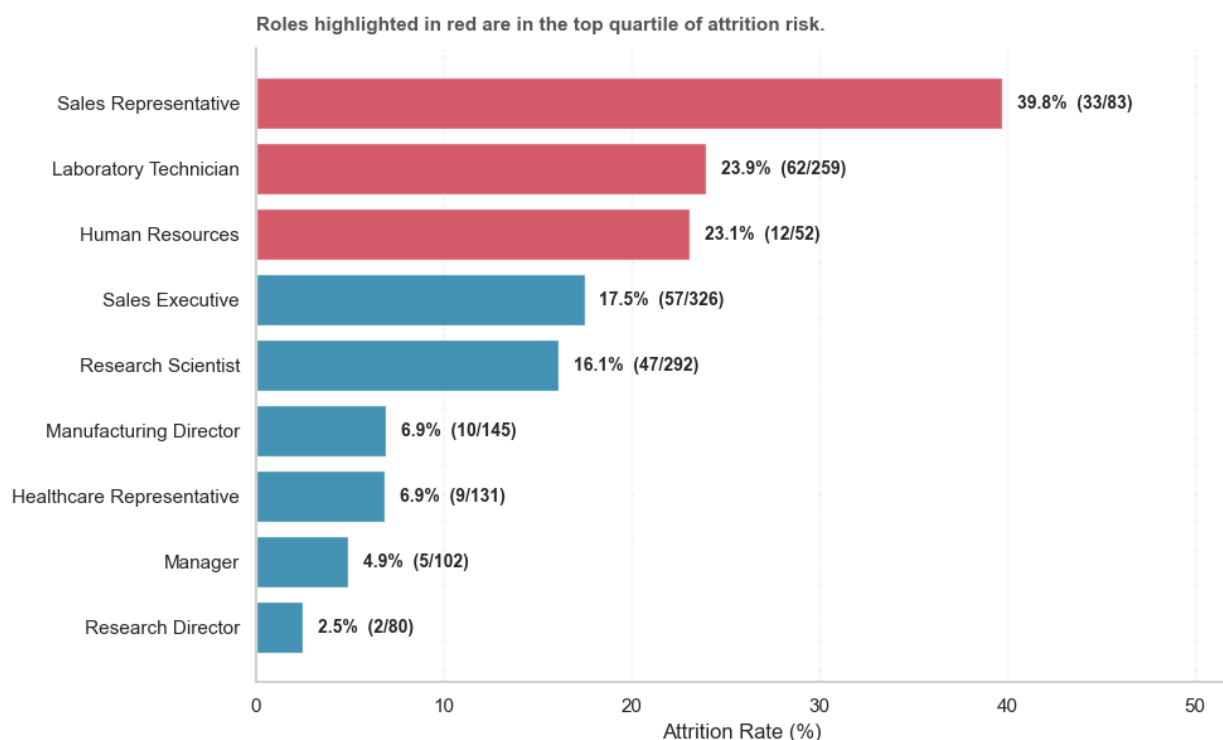


Job role

How it was built

Identifies high-risk roles such as Sales Representative. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Attrition Rate by Job Role



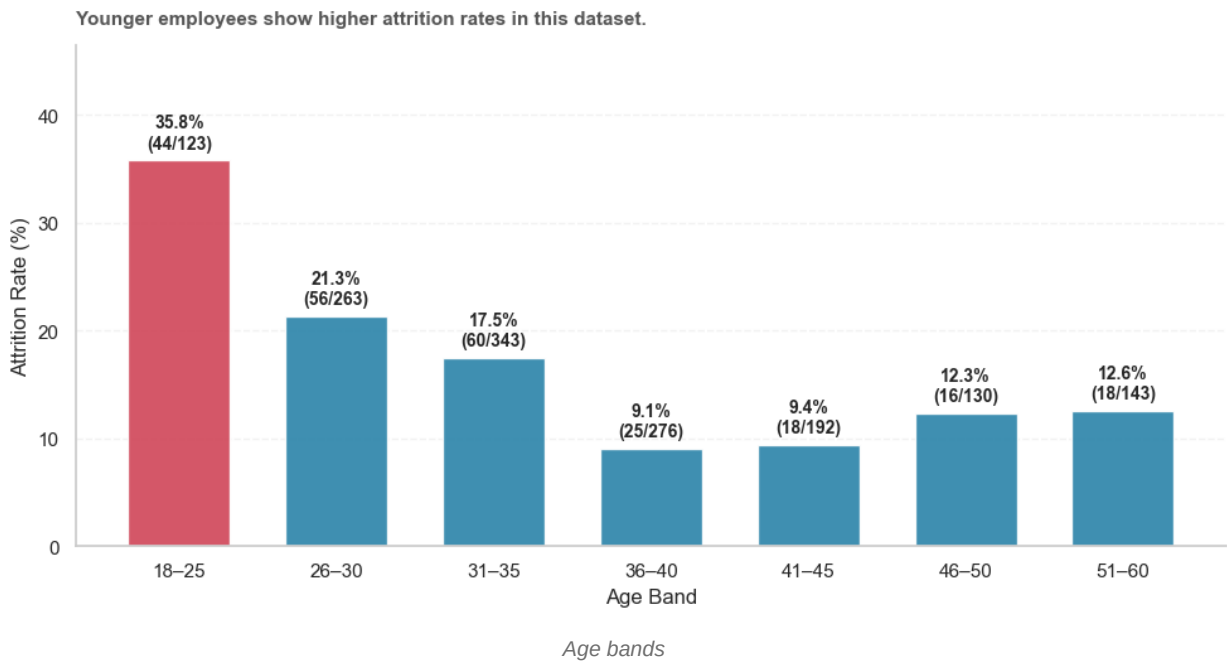
Job role

Age bands

How it was built

Shows younger employees have higher attrition. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Attrition Rate by Age Band

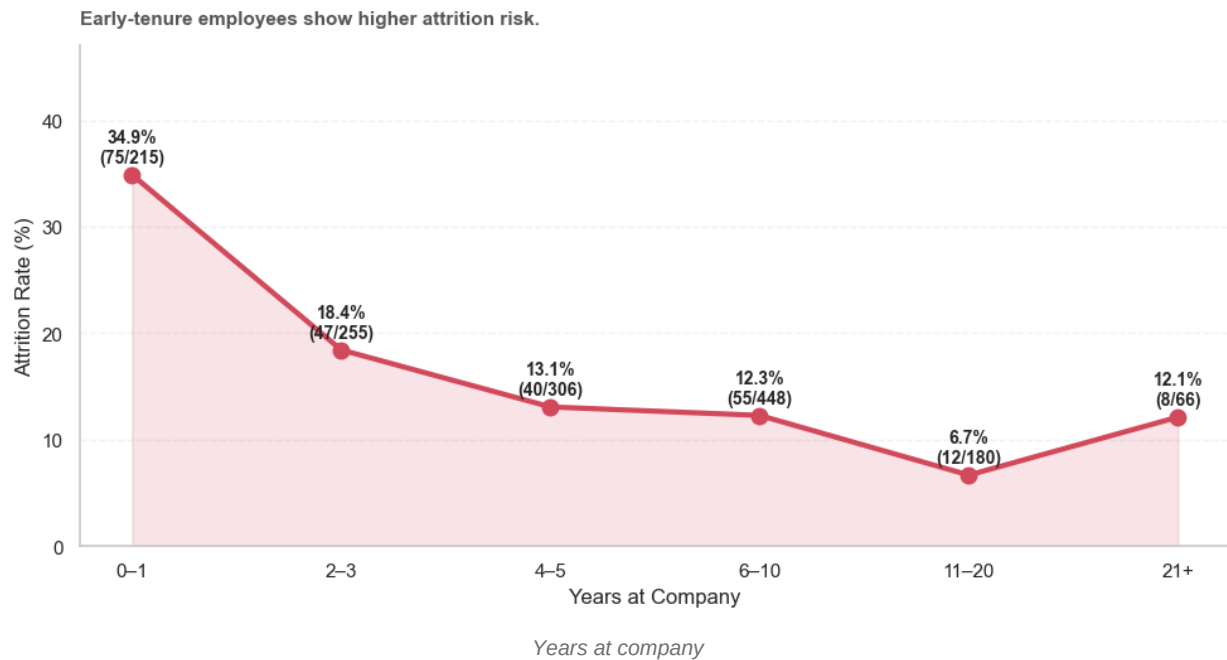


Years at company

How it was built

Shows early-tenure employees have high attrition risk. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Attrition Rate by Years at Company

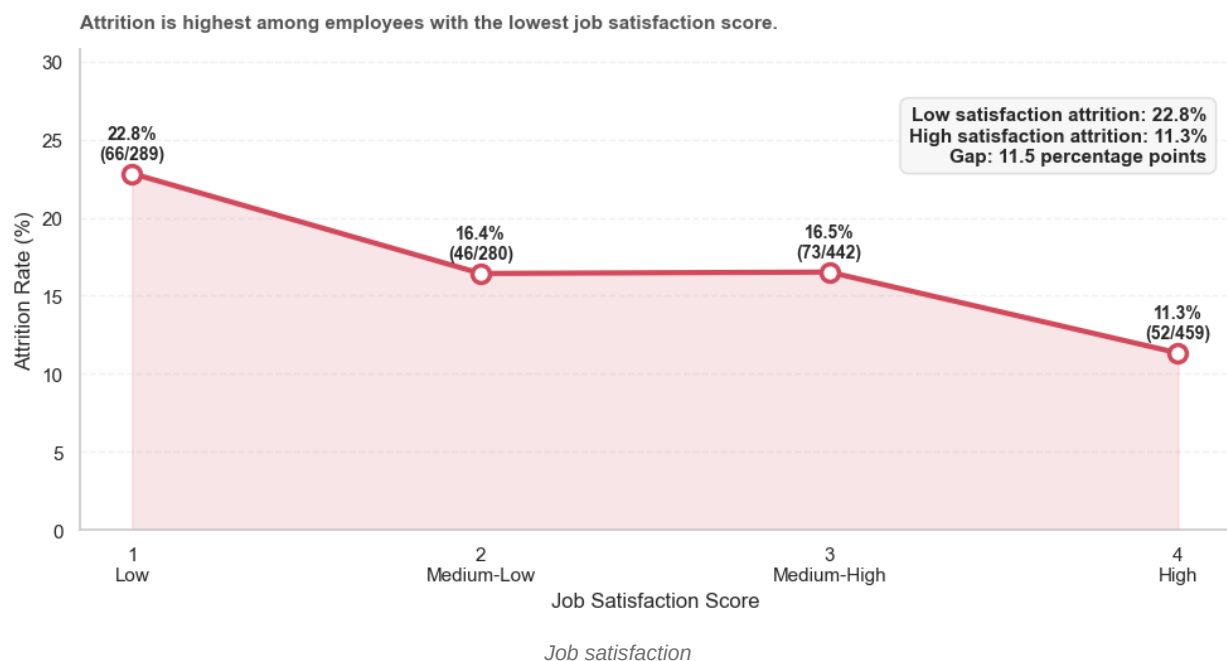


Job satisfaction

How it was built

Shows low satisfaction is associated with higher attrition. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Attrition Rate by Job Satisfaction

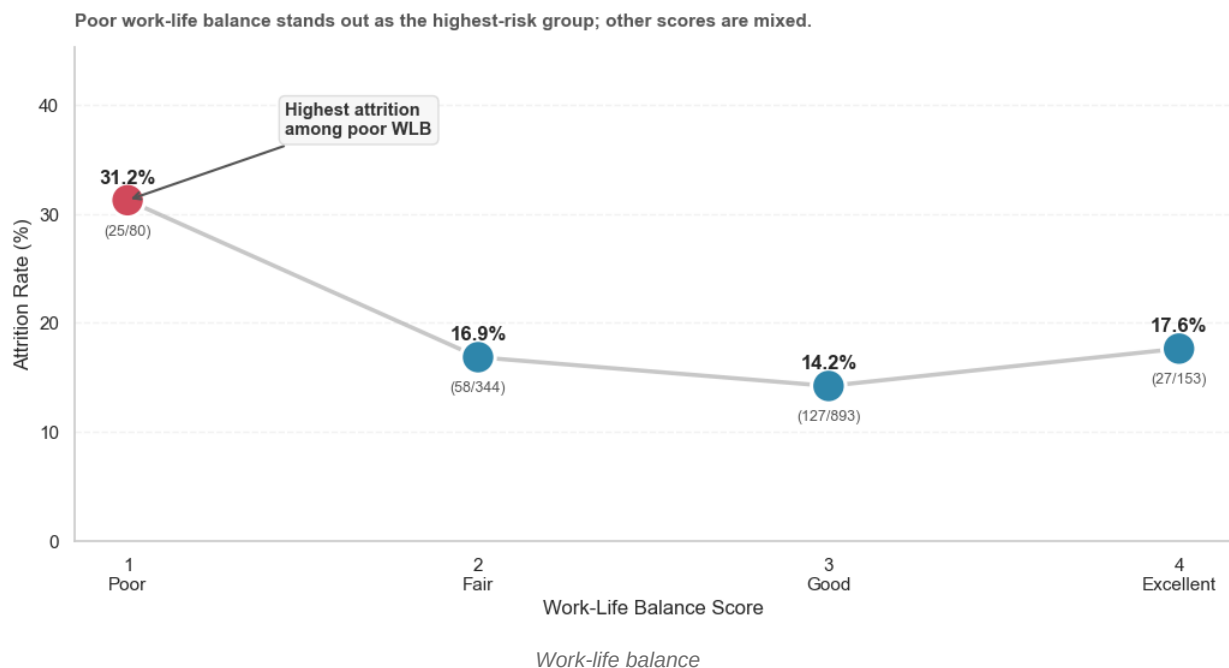


Work-life balance

How it was built

Shows poor work-life balance is a high-risk group. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Attrition Rate by Work-Life Balance

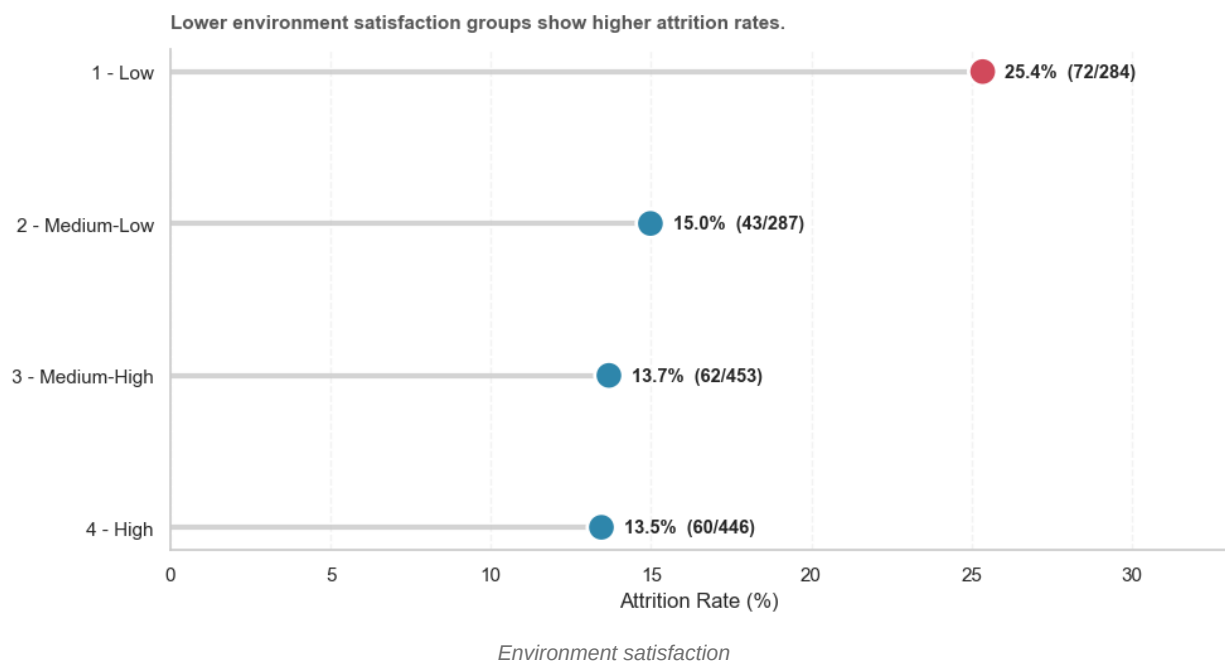


Environment satisfaction

How it was built

Shows low environment satisfaction increases risk. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Attrition Rate by Environment Satisfaction

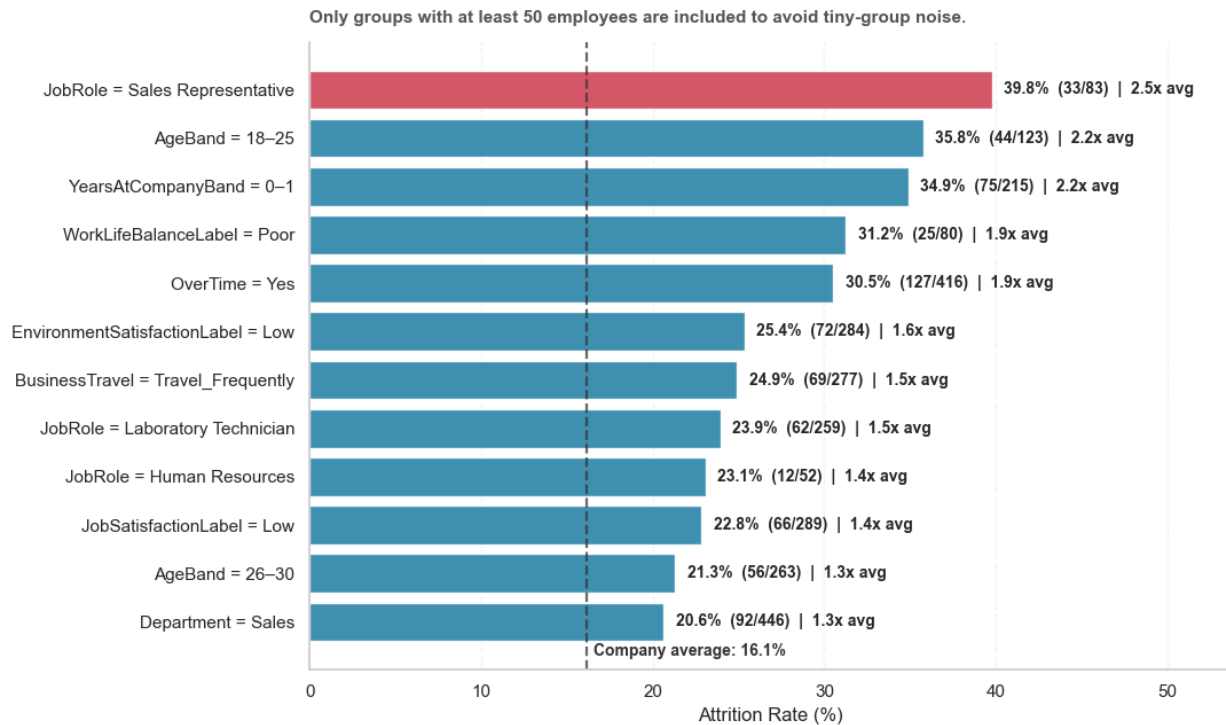


Top risk segments

How it was built

Ranks the highest-risk employee groups. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Top Attrition Risk Segments



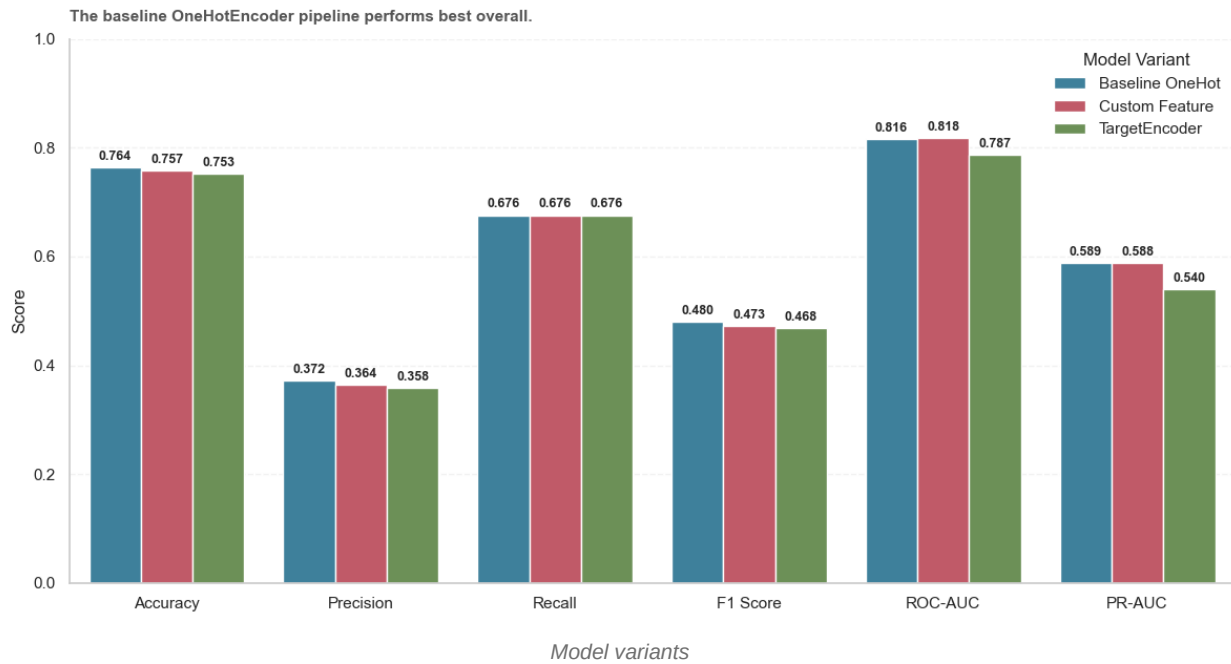
Top risk segments

Model variants

How it was built

Compares baseline, custom feature and TargetEncoder experiments. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Model Performance Comparison Across Preprocessing Strategies

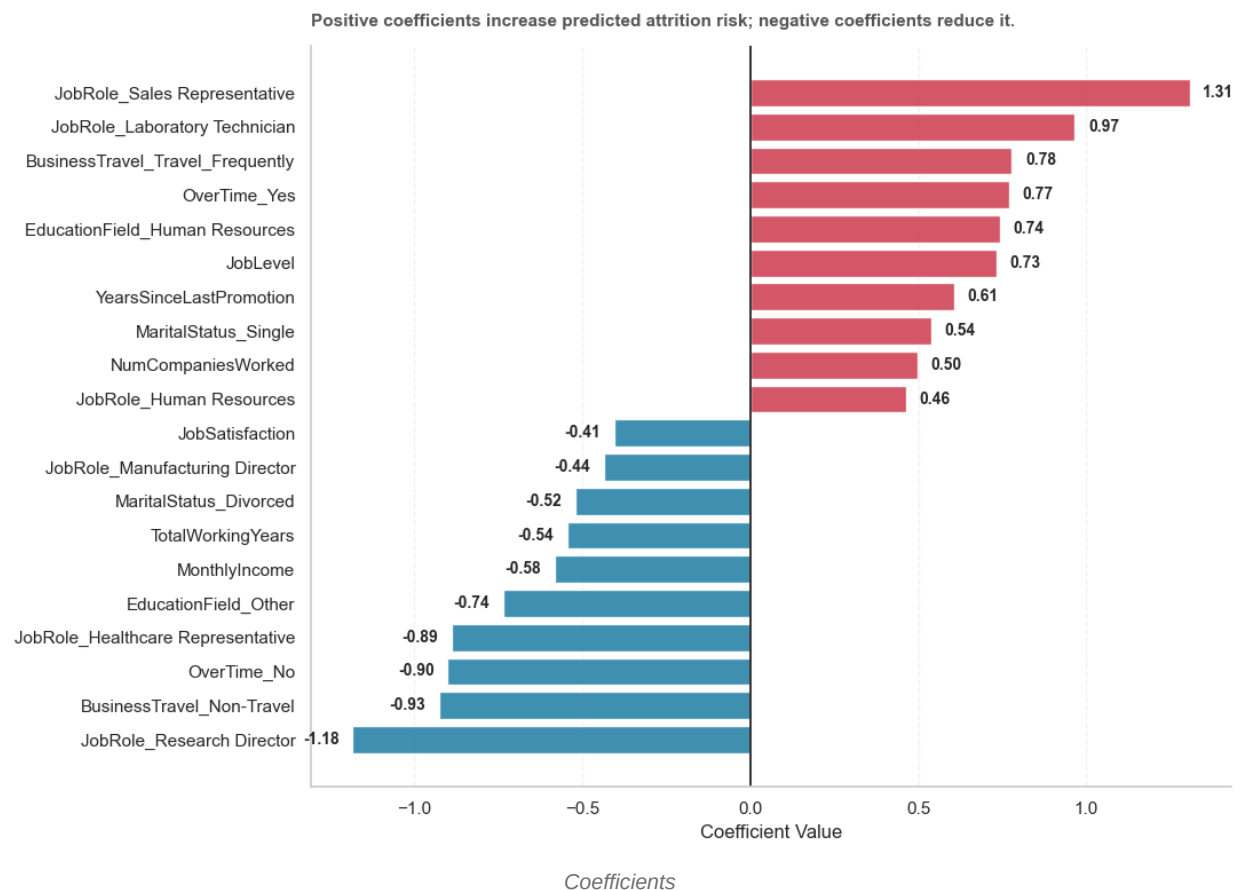


Coefficients

How it was built

Explains which features push predictions toward leave or stay. Most EDA charts used groupby aggregation: group employees -> count employees -> count leavers -> calculate attrition rate -> plot.

Top Logistic Regression Coefficients



SECTION 12

Sources

References used for definitions and documentation.

The project results and charts come from your notebook outputs. The conceptual definitions below are supported by official documentation and public references.

#	Source	What it supports
[1]	scikit-learn StandardScaler documentation	StandardScaler definition and behavior
[2]	scikit-learn MinMaxScaler documentation	MinMaxScaler definition and behavior
[3]	scikit-learn RobustScaler documentation	RobustScaler definition and behavior
[4]	scikit-learn precision_score / recall_score / f1_score documentation	Precision, recall and F1 definitions
[5]	scikit-learn roc_auc_score and average_precision_score documentation	ROC-AUC and PR-AUC definitions
[6]	scikit-learn Pipeline and ColumnTransformer documentation	Pipeline and column-wise preprocessing concepts
[7]	scikit-learn RandomForestClassifier documentation	Random Forest model reference
[8]	XGBoost Python API documentation	XGBClassifier and scale_pos_weight reference
[9]	IBM HR Analytics dataset source as used in the notebook	Dataset context

Source URLs:

<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>

<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html>

<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.RobustScaler.html>

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.precision_score.html

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.recall_score.html

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_auc_score.html

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.average_precision_score.html

<https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.Pipeline.html>

<https://scikit-learn.org/stable/modules/generated/sklearn.compose.ColumnTransformer.html>

<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>

https://xgboost.readthedocs.io/en/stable/python/python_api.html#xgboost.XGBClassifier